1. The nth derivative of y = e^(a*x) with respect to x is:
- **(A)** a * e^(a*x)
- **(B)** a^n * e^(a*x)
- **(C)** n * a * e^(a*x)
- **(D)** e^(a*x) / a^n

2. The nth derivative of y = (a*x + b)^m when m is a positive integer and n > m is:
- **(A)** m! * a^m
- **(B)** 0
- **(C)** m! / (m-n)!
- **(D)** a^n

3. The nth derivative of y = ln(a*x + b) is given by:
- **(A)** (-1)^n * (n-1)! * a^n / (ax+b)^n
- **(B)** (-1)^(n-1) * (n-1)! * a^n / (ax+b)^n
- **(C)** n! * a^n / (ax+b)^n
- **(D)** (-1)^n * n! * a^n / (ax+b)^n

4. The nth derivative of y = sin(a*x + b) is:
- **(A)** a^n * cos(ax + b + n*pi/2)
- **(B)** a^n * sin(ax + b + n*pi/2)
- **(C)** (-1)^n * a^n * sin(ax + b)
- **(D)** a^n * sin(ax + b)

5. The nth derivative of y = cos(a*x + b) is:
- **(A)** a^n * cos(ax + b + n*pi/2)
- **(B)** a^n * sin(ax + b + n*pi/2)
- **(C)** (-a)^n * cos(ax + b)
- **(D)** n * a * cos(ax + b)

6. Leibniz's theorem is used to compute the nth derivative of the:
- **(A)** Sum of two functions
- **(B)** Product of two functions
- **(C)** Quotient of two functions
- **(D)** Composite of two functions

7. In Leibniz's rule d^n(u*v)/dx^n = sum_{r=0}^n C(n,r) * u_(n-r) * v_r, the coefficient C(n,r) is:
- **(A)** n! / (r! * (n-r)!)
- **(B)** n! / r!
- **(C)** r! / n!
- **(D)** 2^n

8. Rolle's theorem requires f(x) to be continuous on [a, b], differentiable on (a, b), and:
- **(A)** f'(a) = f'(b)
- **(B)** f(a) = f(b)
- **(C)** f(a) = -f(b)
- **(D)** f(a) * f(b) < 0

9. Under the conditions of Rolle's theorem, there exists at least one point c in (a, b) such that:
- **(A)** f(c) = 0
- **(B)** f'(c) = 0
- **(C)** f''(c) = 0
- **(D)** f'(c) = 1

10. Lagrange's Mean Value Theorem asserts that for suitable f(x) on [a, b], there is c in (a, b) with f'(c) equal to:
- **(A)** (f(b) + f(a)) / (b - a)
- **(B)** (f(b) - f(a)) / (b - a)
- **(C)** (f(b) - f(a)) / (b + a)
- **(D)** f(b) * f(a)

11. Cauchy's Mean Value Theorem expresses [f(b) - f(a)] / [g(b) - g(a)] as:
- **(A)** f'(c) * g'(c)
- **(B)** f'(c) / g'(c)
- **(C)** g'(c) / f'(c)
- **(D)** f''(c) / g''(c)

12. Maclaurin's series expansion of a function f(x) is a Taylor series expansion centered at:
- **(A)** x = 1
- **(B)** x = 0
- **(C)** x = infinity
- **(D)** x = -1

13. The Maclaurin series for e^x is:
- **(A)** sum x^n / n!
- **(B)** sum (-1)^n * x^n / n!
- **(C)** sum x^(2n) / (2n)!
- **(D)** sum x^n

14. The Maclaurin series for sin(x) contains only:
- **(A)** Even powers of x
- **(B)** Odd powers of x
- **(C)** Constant term
- **(D)** Negative powers of x

15. The Maclaurin series for cos(x) contains only:

(A) Odd powers of x
(B) Even powers of x
(C) Logarithmic terms
(D) Exponential terms

16. Which of the following is an indeterminate form suitable for direct application of L'Hopital's rule?

(A) 0 / infinity
(B) 0 / 0
(C) infinity / 0
(D) 1 / 0

17. The indeterminate form 0 * infinity can be converted into 0/0 or infinity/infinity by writing f*g as:

(A) f + 1/g
(B) f / (1/g)
(C) f * (1/g)
(D) ln(f * g)

18. The indeterminate form 1^infinity is typically evaluated by taking the:

(A) Exponential of the derivative
(B) Natural logarithm of the function
(C) Reciprocal of the base
(D) Square root

19. Curvature kappa of a plane curve measures the rate of turning of the tangent with respect to:

(A) x-coordinate
(B) Arc length s (dpsi / ds)
(C) y-coordinate
(D) Time t

20. The radius of curvature rho is related to curvature kappa by the formula:

(A) rho = kappa^2
(B) rho = 1 / kappa
(C) rho = 2 * pi * kappa
(D) rho = sqrt(kappa)

21. The Cartesian formula for the radius of curvature rho is:

(A) [1 + (y')^2]^(3/2) / |y''|
(B) [1 + (y')^2] / |y''|
(C) [1 + (y')^3] / |y''|
(D) |y''| / [1 + (y')^2]^(3/2)

22. At a point of inflection on a smooth curve, the curvature kappa is:

(A) Infinity
(B) Zero
(C) 1
(D) -1

23. For a straight line, the radius of curvature rho is:

(A) 0
(B) 1
(C) Infinite
(D) Undefined

24. For a circle of radius R, the radius of curvature rho at any point is:

(A) R / 2
(B) R
(C) 2 * R
(D) 1 / R

25. The locus of the centers of curvature of all points on a given curve is called its:

(A) Involute
(B) Evolute
(C) Asymptote
(D) Envelope

26. An asymptote to a curve is a straight line such that the distance between the curve and the line approaches:

(A) Infinity
(B) Zero as the point recedes to infinity
(C) A constant non-zero value
(D) Radius of curvature

27. Asymptotes parallel to the y-axis are found by equating to zero the coefficient of the:

(A) Lowest power of y
(B) Highest power of y
(C) Highest power of x
(D) Lowest power of x

28. Asymptotes parallel to the x-axis are found by equating to zero the coefficient of the:

(A) Highest power of x
(B) Highest power of y
(C) Constant term
(D) Square root of x

29. A curve is symmetric about the x-axis if replacing y by -y leaves the equation:
- **(A)** Reversed
- **(B)** Unchanged
- **(C)** Multiplied by -1
- **(D)** Zero

30. A curve is symmetric about the y-axis if replacing x by -x leaves the equation:
- **(A)** Unchanged
- **(B)** Negative
- **(C)** Inverted
- **(D)** Undefined

31. A curve is symmetric in opposite quadrants (about the origin) if replacing (x, y) by (-x, -y) leaves the equation:
- **(A)** Inverted
- **(B)** Unchanged
- **(C)** Doubled
- **(D)** Zero

32. A curve passes through the origin (0, 0) if the equation has no:
- **(A)** Linear term
- **(B)** Constant term
- **(C)** Quadratic term
- **(D)** Higher order term

33. Tangents at the origin are obtained by equating to zero the:
- **(A)** Highest degree terms
- **(B)** Lowest degree terms
- **(C)** Linear terms only
- **(D)** Constant terms

34. A double point at which two real branches have two distinct tangents is called a:
- **(A)** Cusp
- **(B)** Node
- **(C)** Conjugate point
- **(D)** Point of inflection

35. A double point at which two real branches have two coincident tangents is called a:
- **(A)** Node
- **(B)** Cusp
- **(C)** Isolated point
- **(D)** Inflection point

36. In biotechnology, the Michaelis-Menten rate equation v = Vmax * S / (Km + S) relates reaction velocity to:
- **(A)** Enzyme concentration
- **(B)** Substrate concentration S
- **(C)** Temperature
- **(D)** Bioreactor volume

37. In the Michaelis-Menten model, when S << Km, the reaction rate behaves as:
- **(A)** Zero order
- **(B)** First order with respect to S
- **(C)** Second order
- **(D)** Independent of S

38. In the Michaelis-Menten model, when S >> Km, the reaction rate approaches:
- **(A)** Zero
- **(B)** Vmax (zero order kinetics)
- **(C)** Infinity
- **(D)** Km / 2

39. The maximum specific growth rate mu_max in microbial batch cultures occurs at the point of:
- **(A)** Exponential growth (constant d(ln X)/dt)
- **(B)** Stationary phase
- **(C)** Death phase
- **(D)** Lag phase

40. The inflection point on a sigmoidal cell biomass growth curve X(t) corresponds to:
- **(A)** Maximum biomass concentration
- **(B)** Maximum growth rate dX/dt
- **(C)** Zero growth rate
- **(D)** Cell death

41. The nth derivative of y = x * e^x using Leibniz's rule is:
- **(A)** x * e^x
- **(B)** (x + n) * e^x
- **(C)** n * x * e^x
- **(D)** (x - n) * e^x

42. The nth derivative of y = x^2 * e^(a*x) is given by:
- **(A)** a^n * e^(ax) * [x^2 + 2nx/a + n(n-1)/a^2]
- **(B)** a^n * e^(ax) * [x^2 + nx + n]
- **(C)** a^(n-2) * e^(ax)
- **(D)** x^2 * a^n * e^(ax)

43. If y = sin(m * arcsin(x)), then (1 - x^2)*y2 - x*y1 equals:

(A) -m^2 * y
(B) m^2 * y
(C) 0
(D) m * y

44. If y = (x^2 - 1)^n, then the nth derivative y^(n) is proportional to which classical polynomial?

(A) Hermite polynomial Hn(x)
(B) Legendre polynomial Pn(x)
(C) Laguerre polynomial Ln(x)
(D) Chebyshev polynomial Tn(x)

45. Verify Rolle's theorem for f(x) = x^2 - 4x + 3 on [1, 3]. The point c is:

(A) 1.5
(B) 2.0
(C) 2.5
(D) 0.0

46. For f(x) = x^3 on [0, 2], the point c guaranteed by Lagrange's Mean Value Theorem is:

(A) 2 / sqrt(3)
(B) sqrt(2)
(C) 1.0
(D) 4 / 3

47. Evaluate the limit: lim_{x -> 0} (sin x - x) / x^3:

(A) 1/6
(B) -1/6
(C) 1/3
(D) -1/3

48. Evaluate the limit: lim_{x -> 0} (e^x - 1 - x) / x^2:

(A) 1
(B) 1/2
(C) 0
(D) 2

49. Evaluate the limit: lim_{x -> 0} (tan x - x) / (x - sin x):

(A) 1
(B) 2
(C) -1
(D) 1/2

50. Evaluate the limit: lim_{x -> 0} (1/x - 1/sin x):

(A) 0
(B) 1
(C) Infinity
(D) -1

51. Evaluate the limit: lim_{x -> 0} (cos x)^(1 / x^2):

(A) 1
(B) e^(-1/2)
(C) e^(1/2)
(D) e^(-1)

52. Evaluate the limit: lim_{x -> 0} (1 + 2x)^(1/x):

(A) e
(B) e^2
(C) 2*e
(D) 1

53. Evaluate the limit: lim_{x -> infinity} (1 + a/x)^x:

(A) a
(B) e^a
(C) e^(-a)
(D) ln(a)

54. The radius of curvature of the catenary y = c * cosh(x / c) at any point is:

(A) y^2 / c
(B) c^2 / y
(C) y
(D) c

55. The radius of curvature of the cycloid x = a(theta - sin theta), y = a(1 - cos theta) is:

(A) 2a * sin(theta / 2)
(B) 4a * sin(theta / 2)
(C) 4a * cos(theta / 2)
(D) a * sin(theta)

56. The radius of curvature of the parabola y^2 = 4ax at the vertex (0, 0) is:

(A) a
(B) 2a
(C) 4a
(D) a / 2

57. The pedal equation for an ellipse is 1/p^2 = 1/a^2 + 1/b^2 - r^2/(a^2*b^2). The formula for rho in pedal coordinates is:

(A) r * (dr / dp)
(B) p * (dp / dr)
(C) r^2 * (dr / dp)
(D) dr / dp

58. For the curve r = a * (1 + cos theta) (cardioid), the radius of curvature rho is:

(A) (4/3) * a * cos(theta / 2)
(B) (2/3) * a * sqrt(2*a*r)
(C) (2/3) * sqrt(2*a*r)
(D) a * sin(theta)

59. The oblique asymptotes of the curve x^3 + y^3 = 3axy (Folium of Descartes) are given by:

(A) x + y = 0
(B) x + y + a = 0
(C) x - y + a = 0
(D) y = x

60. The vertical asymptotes of the rational function f(x) = (x^2 + 1) / (x^2 - 4) are:

(A) x = 1, x = -1
(B) x = 2, x = -2
(C) y = 1
(D) y = 4

61. The horizontal asymptote of the rational function f(x) = (3x^2 + 2) / (x^2 - 5) is:

(A) y = 0
(B) y = 3
(C) y = -5
(D) x = 3

62. In curve tracing, the curve y^2 = x^2 * (a + x) / (a - x) has a loop in the region:

(A) -a <= x <= 0
(B) 0 <= x < a
(C) x >= a
(D) x <= -a

63. The curve y^2 = x^3 (semicubical parabola) has at the origin (0, 0) a:

(A) Node
(B) Cusp of the first kind
(C) Conjugate point
(D) Point of inflection

64. The curve y^2 = x^2 * (x + 1) has at the origin (0, 0) a:

(A) Cusp
(B) Node (two real distinct tangents y = +-x)
(C) Isolated point
(D) Smooth minimum

65. Taylor's theorem with remainder states that f(a+h) = f(a) + h*f'(a) + ... + (h^n / n!)*f^(n)(a + theta*h), where theta lies in:

(A) (0, 1)
(B) (1, 2)
(C) (-1, 1)
(D) [a, b]

66. The coefficient of x^3 in the Maclaurin expansion of tan(x) is:

(A) 1/3
(B) -1/3
(C) 2/3
(D) 1/6

67. The coefficient of x^4 in the Maclaurin expansion of ln(1 + x) is:

(A) 1/4
(B) -1/4
(C) 1/24
(D) -1/12

68. If an enzyme-catalyzed reaction has Vmax = 50 umol/(L*min) and Km = 2 mmol/L, what is the rate v at substrate S = 2 mmol/L?

(A) 50
(B) 25
(C) 12.5
(D) 100

69. In a Lineweaver-Burk double reciprocal plot (1/v vs 1/S), the y-intercept represents:

(A) Km
(B) 1 / Vmax
(C) -1 / Km
(D) Vmax / Km

70. In a Lineweaver-Burk plot, the x-intercept is equal to:

(A) -1 / Km
(B) 1 / Km
(C) -Km
(D) 1 / Vmax

71. The slope of the Lineweaver-Burk linear plot is given by:

(A) Km / Vmax
(B) Vmax / Km
(C) 1 / (Km * Vmax)
(D) Km * Vmax

72. For logistic population growth N(t) = K / (1 + A * e^(-r*t)), the population at the inflection point is:

(A) K
(B) K / 2
(C) K / 4
(D) K / 3

73. The radius of curvature of the curve y = ln(sec x) at x = 0 is:

(A) 0
(B) 1
(C) 1/2
(D) Infinity

74. The coordinates (x_bar, y_bar) of the center of curvature for curve y = f(x) are given by x_bar = x - y1*(1+y1^2)/y2 and y_bar equal to:

(A) y + (1 + y1^2) / y2
(B) y - (1 + y1^2) / y2
(C) y + y2 / (1 + y1^2)
(D) y * y1 / y2

75. The evolute of the parabola y^2 = 4ax is the semicubical parabola:

(A) 27ay^2 = 4(x - 2a)^3
(B) 27ax^2 = 4(y - 2a)^3
(C) y^2 = 4a(x - a)
(D) x^2 + y^2 = 4a^2

76. If y = (x + sqrt(1 + x^2))^m, find the value of (1 + x^2)*y_{n+2} + (2n+1)*x*y_{n+1} + (n^2 - m^2)*y_n:

(A) m^2
(B) 0
(C) -m^2
(D) n^2

77. If y = e^(m * arcsin x), what is the coefficient of y_n in the expansion of (1 - x^2)*y_{n+2} - (2n+1)*x*y_{n+1}?

(A) -(n^2 + m^2)
(B) (n^2 + m^2)
(C) m^2 - n^2
(D) n^2 - m^2

78. If y = cos(m * ln x), evaluate x^2 * y_{n+2} + (2n+1)*x * y_{n+1} + (n^2 + m^2)*y_n:

(A) 1
(B) 0
(C) -1
(D) m^2

79. Find the nth derivative of y = x / (x^2 - a^2) at x = 0 when n is even:

(A) 0
(B) n! / a^(n+1)
(C) (-1)^n * n!
(D) n * a^n

80. Evaluate: lim_{x -> 0} (sin x / x)^(1 / x^2):

(A) e^(-1/6)
(B) e^(1/6)
(C) e^(-1/3)
(D) 1

81. Evaluate: lim_{x -> 0} [(1/x^2) - (cot x)^2]:

(A) 1/3
(B) 2/3
(C) 1/2
(D) 0

82. Evaluate: lim_{x -> 0} (e^x * sin x - x * (1 + x)) / x^3:

(A) 1/6
(B) 1/3
(C) -1/6
(D) 1/2

83. Find the radius of curvature of the astroid x^(2/3) + y^(2/3) = a^(2/3) at any point:

(A) 3 * (a * x * y)^(1/3)
(B) 3a * sin(2*theta)
(C) 3 * a^(2/3)
(D) a / 3

84. The center of curvature of the parabola y^2 = 4ax at the point (at^2, 2at) has coordinates (x_bar, y_bar):

(A) (2a + 3at^2, -2at^3)
(B) (a + 2at^2, -at^3)
(C) (3at^2, -at^3)
(D) (2a, 0)

85. Find all asymptotes of the cubic curve x^3 + 2x^2*y - x*y^2 - 2y^3 + 4x^2 - 4y^2 - 1 = 0:

(A) x + y = 0, x - y = 0, x + 2y = 0
(B) x - y + 2 = 0, x + y + 2 = 0, x + 2y = 0
(C) x + y = 1, x - y = 1
(D) y = 2x, y = -x

86. The radius of curvature of the curve r^n = a^n * cos(n * theta) is given by:

(A) a^n / ((n+1) * r^(n-1))
(B) a^n / (n * r^(n-1))
(C) r^(n+1) / a^n
(D) (n+1) * a^n / r

87. For the cardioid r = a*(1 - cos theta), the radius of curvature at theta = pi is:

(A) 4a / 3
(B) 8a / 3
(C) 2a / 3
(D) 2a

88. Trace the lemniscate r^2 = a^2 * cos(2*theta). The curve exists only in the angular intervals:

(A) [-pi/4, pi/4] and [3pi/4, 5pi/4]
(B) [0, pi/2]
(C) [pi/4, 3pi/4]
(D) [0, 2pi]

89. The maximum value of the function f(x) = x * (1 - x)^2 on [0, 1] occurs at x equal to:

(A) 1/2
(B) 1/3
(C) 2/3
(D) 1/4

90. In a fed-batch bioreactor, cell growth follows the Monod equation mu = mu_max * S / (Ks + S). If mu_max = 0.8 h^-1 and Ks = 0.5 g/L, at what substrate concentration S is mu = 0.6 h^-1?

(A) 1.0 g/L
(B) 1.5 g/L
(C) 2.0 g/L
(D) 0.75 g/L

91. The expansion of e^(x * sin x) in ascending powers of x up to x^4 is:

(A) 1 + x^2 - x^4/3
(B) 1 + x^2 + x^4/3
(C) 1 + x^2 + x^4/6
(D) 1 - x^2 + x^4/2

92. Evaluate: lim_{x -> 0} (arcsin x - x) / (x^3):

(A) 1/6
(B) -1/6
(C) 1/3
(D) 1/2

93. If f(x) = x*(x-1)*(x-2)*(x-3), how many real roots does f'(x) have?

(A) 1
(B) 2
(C) 3 (all in (0, 3))
(D) 4

94. The envelope of the family of straight lines x*cos(alpha) + y*sin(alpha) = p (where p is constant) is:

(A) A circle x^2 + y^2 = p^2
(B) A parabola y^2 = 4px
(C) An ellipse
(D) A hyperbola

95. The radius of curvature of the ellipse x^2/a^2 + y^2/b^2 = 1 at the end of the major axis (a, 0) is:

(A) b^2 / a
(B) a^2 / b
(C) a * b
(D) sqrt(a * b)

96. The radius of curvature of the ellipse x^2/a^2 + y^2/b^2 = 1 at the end of the minor axis (0, b) is:

(A) b^2 / a
(B) a^2 / b
(C) a / b
(D) b / a

97. Evaluate the limit: lim_{x -> 0} (1/x^2 - 1/(tan x)^2):

(A) 1/3
(B) 2/3
(C) 1
(D) 0

98. In enzyme inhibition kinetics, competitive inhibition increases the apparent Km while leaving Vmax:

(A) Doubled
(B) Halved
(C) Unchanged
(D) Zero

99. In non-competitive enzyme inhibition, the inhibitor binds to both free enzyme and enzyme-substrate complex, causing:

(A) Vmax to decrease while Km remains unchanged
(B) Km to increase while Vmax remains unchanged
(C) Both Vmax and Km to double
(D) Reaction to stop instantly

100. The chord of curvature through the origin for the cardioid r = a*(1 + cos theta) has length:

(A) 4r / 3
(B) 2r / 3
(C) r / 3
(D) 2r

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. State Leibniz's Theorem for the nth derivative of the product of two functions and prove it rigorously by mathematical induction.

Q2. If y = (arcsin x)^2, prove that: (1 - x^2) * y2 - x * y1 - 2 = 0. Hence, using Leibniz's theorem, establish that: (1 - x^2) * y_{n+2} - (2n + 1) * x * y_{n+1} - n^2 * y_n = 0.

Q3. Derive the general formula for the nth derivative of y = e^(a*x) * cos(b*x + c). Hence, compute the 5th derivative of y = e^(2x) * cos(3x + 1).

Q4. If y = arctan(x), prove that (1 + x^2) * y_{n+2} + 2(n + 1) * x * y_{n+1} + n(n + 1) * y_n = 0. Hence determine the value of y_n(0) for all positive integers n.

Q5. State Rolle's Theorem, give its geometrical interpretation, and provide a rigorous proof based on the Extreme Value Theorem.

Q6. State and prove Lagrange's Mean Value Theorem (LMVT). Use it to prove that for all x > 0: x / (1 + x) < ln(1 + x) < x.

Q7. State and prove Cauchy's Mean Value Theorem (CMVT). Deduce Lagrange's Mean Value Theorem as a particular case.

Q8. Derive Taylor's Theorem with Lagrange's and Cauchy's forms of remainder for a function of one variable.

Q9. Expand ln(1 + x) and e^x in Maclaurin series with remainder, state their intervals of convergence, and estimate the error when approximating e by its first 5 terms.

Q10. Prove L'Hopital's Rule for the indeterminate form 0/0 using Cauchy's Mean Value Theorem. Evaluate: lim_{x -> 0} [(e^x - e^(-x) - 2x) / (x - sin x)].

Q11. Evaluate the following indeterminate limits: (i) lim_{x -> 0} [(tan x - x) / x^3], (ii) lim_{x -> 0} [(1 / x^2) - (1 / sin^2 x)].

Q12. Evaluate the exponential indeterminate limits: (i) lim_{x -> 0} (sin x / x)^(1 / x^2) [Form 1^infinity], (ii) lim_{x -> 0+} x^(sin x) [Form 0^0].

Q13. Define curvature kappa and radius of curvature rho. Starting from the intrinsic definition kappa = dpsi / ds, rigorously derive the Cartesian formula: rho = [1 + (y')^2]^(3/2) / |y''|.

Q14. Derive the formula for the radius of curvature in parametric form: rho = (x'^2 + y'^2)^(3/2) / |x' * y'' - y' * x''|. Find the radius of curvature for the cycloid x = a(theta - sin theta), y = a(1 - cos theta) at any point theta.

Q15. Derive the polar coordinate formula for radius of curvature: rho = (r^2 + r'^2)^(3/2) / |r^2 + 2*r'^2 - r * r''|. Find rho for the cardioid r = a*(1 + cos theta).

Q16. Derive the pedal formula for the radius of curvature: rho = r * (dr / dp). Apply this formula to find the radius of curvature of the equiangular spiral r = a * e^(theta * cot alpha).

Q17. Derive the coordinates (x_bar, y_bar) of the center of curvature for a curve y = f(x). Hence, find the equation of the circle of curvature at any point.

Q18. Define the evolute of a curve. Derive the parametric coordinates of the center of curvature for the parabola y^2 = 4ax at point (at^2, 2at), and eliminate t to find the Cartesian equation of its evolute.

Q19. Define an asymptote of an algebraic curve. Explain the systematic method for finding asymptotes parallel to coordinate axes and oblique asymptotes y = mx + c.

Q20. Find all the asymptotes (parallel and oblique) of the cubic curve: x^3 + 3x^2*y - 4y^3 - x + y + 3 = 0.

Q21. Determine all asymptotes of the Folium of Descartes: x^3 + y^3 = 3axy. Explain how the curve approaches its asymptote.

Q22. Classify double points (singular points) of algebraic curves. Derive the conditions for nodes, cusps, and conjugate points, and examine the origin for the curve y^2 = x^2 * (x + 1).

Q23. Describe the systematic 6-step algorithm for tracing Cartesian curves: (1) Symmetry, (2) Origin and tangents, (3) Intercepts, (4) Asymptotes, (5) Regions of existence, (6) Sign of derivatives.

Q24. Apply the curve tracing algorithm to thoroughly analyze and sketch the Cissoid of Diocles: y^2 * (2a - x) = x^3 (where a > 0).

Q25. Apply the curve tracing algorithm to analyze the Strophoid: y^2 * (a - x) = x^2 * (a + x) (where a > 0).

Q26. Trace the polar curve: Cardioid r = a * (1 + cos theta) (where a > 0).

Q27. Formulate the First and Second Derivative Tests for local extrema and points of inflection of a twice-differentiable function f(x). Prove the Second Derivative Test.

Q28. Derive the Michaelis-Menten enzyme kinetics rate equation v = Vmax * S / (Km + S) using the Briggs-Haldane quasi-steady-state differential formulation.

Q29. Model microbial cell growth using the logistic differential equation dX/dt = r * X * (1 - X / K). Solve this non-linear ODE by separation of variables, find the inflection point, and determine when growth velocity is maximized.

Q30. Analyze substrate inhibition enzyme kinetics governed by the Haldane model: v = Vmax * S / (Km + S + S^2 / Ki). Determine the optimal substrate concentration S_opt that maximizes reaction velocity and compute the maximum inhibited velocity v_opt.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[B]	**Q21:**	[A]	**Q41:**	[B]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[B]
Q3:	[B]	**Q23:**	[C]	**Q43:**	[A]	**Q63:**	[B]	**Q83:**	[A]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[B]	**Q64:**	[B]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[B]	**Q45:**	[B]	**Q65:**	[A]	**Q85:**	[B]
Q6:	[B]	**Q26:**	[B]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[B]	**Q47:**	[B]	**Q67:**	[B]	**Q87:**	[B]
Q8:	[B]	**Q28:**	[A]	**Q48:**	[B]	**Q68:**	[B]	**Q88:**	[A]
Q9:	[B]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[B]	**Q89:**	[B]
Q10:	[B]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[B]
Q11:	[B]	**Q31:**	[B]	**Q51:**	[B]	**Q71:**	[A]	**Q91:**	[C]
Q12:	[B]	**Q32:**	[B]	**Q52:**	[B]	**Q72:**	[B]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[B]	**Q53:**	[B]	**Q73:**	[B]	**Q93:**	[C]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[B]	**Q35:**	[B]	**Q55:**	[B]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[B]	**Q36:**	[B]	**Q56:**	[B]	**Q76:**	[B]	**Q96:**	[B]
Q17:	[B]	**Q37:**	[B]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[B]
Q18:	[B]	**Q38:**	[B]	**Q58:**	[C]	**Q78:**	[B]	**Q98:**	[C]
Q19:	[B]	**Q39:**	[A]	**Q59:**	[B]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

Step 3: Differentiate implicitly with respect to x:
d/dx [(1 - x^2) * (y1)^2] = d/dx [4 * y]
=> (1 - x^2) * [2 * y1 * y2] + (-2x) * (y1)^2 = 4 * y1.
Divide the entire equation by 2 * y1 (assuming y1 != 0):
(1 - x^2) * y2 - x * y1 = 2
=> (1 - x^2) * y2 - x * y1 - 2 = 0. (First result established).
Step 4: Differentiate n times using Leibniz's Theorem:
Apply d^n/dx^n to each term:
Term 1: d^n / dx^n [(1 - x^2) * y2]:
Here u = y2, v = 1 - x^2 (v' = -2x, v'' = -2, v''' = 0).
= C(n, 0) * y_{n+2} * (1 - x^2) + C(n, 1) * y_{n+1} * (-2x) + C(n, 2) * y_n * (-2)
= (1 - x^2) * y_{n+2} + n * (-2x) * y_{n+1} + [n(n - 1) / 2] * (-2) * y_n
= (1 - x^2) * y_{n+2} - 2n * x * y_{n+1} - n(n - 1) * y_n.
Term 2: d^n / dx^n [-x * y1]:
Here u = y1, v = -x (v' = -1, v'' = 0).
= - [C(n, 0) * y_{n+1} * x + C(n, 1) * y_n * 1]
= -x * y_{n+1} - n * y_n.
Term 3: d^n / dx^n [-2] = 0 (constant).
Step 5: Combine all terms:
[(1 - x^2) * y_{n+2} - 2n * x * y_{n+1} - n(n - 1) * y_n] - [x * y_{n+1} + n * y_n] = 0
=> (1 - x^2) * y_{n+2} - (2n + 1) * x * y_{n+1} - [n(n - 1) + n] * y_n = 0
=> (1 - x^2) * y_{n+2} - (2n + 1) * x * y_{n+1} - [n^2 - n + n] * y_n = 0
=> (1 - x^2) * y_{n+2} - (2n + 1) * x * y_{n+1} - n^2 * y_n = 0.
Hence proved.

>>Solution to Question 3: Derive the general formula for the nth derivative of y = e^(ax) * cos(bx + c). Hence, compute the 5th derivative of y = e^(2x) * cos(3x +

- Expression for 5th derivative:
y_5 = 13^(5/2) * e^(2x) * cos(3x + 1 + 5 * arctan(1.5))
= 169 * sqrt(13) * e^(2x) * cos(3x + 1 + 5 * arctan(1.5)).

>>Solution Statement: Rolle's Theorem: Let f(x) be a real-valued function defined on the interval [a, b] such that: 1. f(x) is continuous on [a, b] 2. f(x) is differentiable on (a, b) 3. f(a) = f(b) Then there exists at least one c in (a, b) such that f'(c) = 0. In other words, if a function satisfies these conditions, there must be at least one point between a and b where the derivative of the function is zero. Proof: To prove Rolle's Theorem, we will use the Extreme Value Theorem and the definition of a derivative. Since f(x) is continuous on [a, b], it must attain its maximum and minimum values on this interval. Let M be the maximum value and m be the minimum value of f(x) on [a, b]. If M = m, then f(x) is constant on [a, b], and f'(x) = 0 for all x in (a, b). In this case, any c in (a, b) will satisfy the theorem. If M > m, then at least one of M or m must be attained at a point c in (a, b), since f(a) = f(b). Without loss of generality, assume that M is attained at c. Then, for all x in (a, b), we have f(x) ≤ f(c). This implies that the derivative of f(x) at c is zero, since the slope of the tangent line to the graph of f(x) at c is horizontal. Therefore, f'(c) = 0, and the theorem is proved.

2. Geometrical Interpretation:
- The graph of y = f(x) connects points A(a, f(a)) and B(b, f(b)) where both endpoints have identical heights (f(a) = f(b)).
- The theorem asserts that if the curve is smooth (no sharp corners) and unbroken, there must be at least one point (c, f(c)) between A and B where the tangent line is horizontal (parallel to the x-axis).
3. Rigorous Analytical Proof:
- Since f(x) is continuous on the closed bounded interval [a, b], by the Weierstrass Extreme Value Theorem, f attains its global maximum M and global minimum m on [a, b]:
M = max_{x in [a, b]} f(x), m = min_{x in [a, b]} f(x).
- Case 1: M = m.
Then f(x) is a constant function: f(x) = M for all x in [a, b].
The derivative of a constant function is zero everywhere: f'(x) = 0 for all x in (a, b).
Thus any point c in (a, b) satisfies f'(c) = 0.
- Case 2: M != m.
Since f(a) = f(b), at least one of M or m must be attained at an interior point c in (a, b).
Without loss of generality, assume the maximum M is attained at c in (a, b), so f(c) >= f(x) for all x in [a, b].
Since f is differentiable on (a, b), f'(c) exists and equals both one-sided limits:
* Left-hand derivative: For h < 0 (sufficiently small):
f(c + h) - f(c) <= 0 => [f(c + h) - f(c)] / h >= 0 (since h < 0).
Taking limit h -> 0-: f'(c) >= 0.
* Right-hand derivative: For h > 0 (sufficiently small):
f(c + h) - f(c) <= 0 => [f(c + h) - f(c)] / h <= 0 (since h > 0).
Taking limit h -> 0+: f'(c) <= 0.
Since f'(c) exists:
0 <= f'(c) <= 0 => f'(c) = 0.
- Therefore, there exists c in (a, b) such that f'(c) = 0. (Hence proved).

>> Solution to Question 8: State and prove Lagrange's Mean Value Theorem (LMVT). Use it to prove that for all x > 0: x / (1 + x) < ln(1 + x) < x.

=> ln(1 + x) / x = 1 / (1 + c).
- Since 0 < c < x:
1 < 1 + c < 1 + x.
- Inverting the inequality (all terms positive):
1 / (1 + x) < 1 / (1 + c) < 1.
- Substitute 1 / (1 + c) = ln(1 + x) / x:
1 / (1 + x) < ln(1 + x) / x < 1.
- Multiply throughout by x > 0:
x / (1 + x) < ln(1 + x) < x. (Proved).

>> Solution: Question: Taylor's Theorem with Lagrange's Remainder: Mean Value Theorem: Calculus: Used to express a function as an infinite sum of terms: Mean Value Theorem: Calculus: Describes the behavior of a function over an interval: Lagrange's Remainder: Taylor's Theorem: Calculus: Provides an estimate of the error in a particular case.

1. Statement of Taylor's Theorem:
Let f: [a, a + h] -> R be such that f, f', f'', ..., f^(n-1) are continuous on [a, a + h] and f^(n) exists on (a, a + h).
Then:
f(a + h) = f(a) + h * f'(a) + (h^2 / 2!) * f''(a) + ... + [h^(n-1) / (n - 1)!] * f^(n-1)(a) + R_n,
where R_n is the Remainder term after n terms.
2. Derivation using Auxiliary Function:
- For x in [a, a + h], define:
phi(x) = f(a + h) - [f(x) + (a + h - x)*f'(x) + ((a + h - x)^2 / 2!)*f''(x) + ... + ((a + h - x)^(n-1) / (n-1)!)*f^(n-1)(x)].
- Note that at x = a + h, all terms vanish except f(a + h) - f(a + h) = 0, so phi(a + h) = 0.
- At x = a, phi(a) = R_n.
- Differentiate phi(x) with respect to x (terms telescope pairwise):
phi'(x) = -f'(x) - [-f'(x) + (a+h-x)*f''(x)] - [-(a+h-x)*f''(x) + ((a+h-x)^2/2!)*f'''(x)] -
...
= - [(a + h - x)^(n-1) / (n - 1)!] * f^(n)(x).
- Now choose another function psi(x) = (a + h - x)^p, where p >= 1 is an integer.
Then psi(a + h) = 0, psi(a) = h^p, and psi'(x) = -p * (a + h - x)^(p - 1).
- Apply Cauchy's Mean Value Theorem to phi(x) and psi(x) on [a, a + h]:
[phi(a + h) - phi(a)] / [psi(a + h) - psi(a)] = phi'(c) / psi'(c) for some c in (a, a + h).
[0 - R_n] / [0 - h^p] = R_n / h^p = [- ((a + h - c)^(n-1) / (n - 1)!) * f^(n)(c)] / [-p * (a + h - c)^(p - 1)]
=> R_n = [h^p / (p * (n - 1)!)] * (a + h - c)^(n - p) * f^(n)(c).
- Write c = a + theta * h, where 0 < theta < 1. Then a + h - c = h * (1 - theta).
General Schlomilch-Roche Remainder:
R_n = [h^n * (1 - theta)^(n - p) / (p * (n - 1)!)] * f^(n)(a + theta * h).
3. Deduction of Specific Remainder Forms:
- Lagrange's Form of Remainder (set p = n):
R_n = [h^n / n!] * f^(n)(a + theta * h), where 0 < theta < 1.
- Cauchy's Form of Remainder (set p = 1):
R_n = [h^n * (1 - theta)^(n - 1) / (n - 1)!] * f^(n)(a + theta * h), where 0 < theta < 1.

>> Solution to Question 9: Expand ln(1 + x) and e^x in Maclaurin series with remainder, state their intervals of convergence, and estimate the error when approxim

- f(x) = ln(1 + x) => f(0) = 0.
- f'(x) = (1 + x)^-1 => f'(0) = 1.
- f''(x) = -(1 + x)^-2 => f''(0) = -1.
- f'''(x) = 2!(1 + x)^-3 => f'''(0) = 2!.
- In general: f^(n)(0) = (-1)^(n - 1) * (n - 1)!.
- Maclaurin series:

ln(1 + x) = x - x^2/2 + x^3/3 - x^4/4 + ... + (-1)^(n - 1) * (x^n / n) + ...

- Interval of Convergence:

Ratio test: lim |a_{n+1} / a_n| = |x| < 1.

At x = 1: 1 - 1/2 + 1/3 - 1/4 + ... (converges by alternating series test to ln 2).

At x = -1: -(1 + 1/2 + 1/3 + ...) (diverges, harmonic series).

Interval of convergence is: -1 < x <= 1.

- Method 1: Using Maclaurin series of tan x:
tan x = x + x^3/3 + 2x^5/15 + O(x^7).
tan x - x = x^3/3 + 2x^5/15 + ...
Therefore:
L1 = lim_{x -> 0} [(x^3/3 + 2x^5/15 + ...) / x^3] = lim_{x -> 0} [1/3 + 2x^2/15 + ...] = 1/3.
- Method 2: By L'Hopital's Rule:
L1 = lim_{x -> 0} [(sec^2 x - 1) / (3x^2)] = lim_{x -> 0} [tan^2 x / (3x^2)]
= (1/3) * [lim_{x -> 0} (tan x / x)]^2 = (1/3) * (1)^2 = 1/3.
2. Part (ii): Evaluate L2 = lim_{x -> 0} [(1 / x^2) - (1 / sin^2 x)] (Form infinity - infinity):
- Combine into a single fraction:
L2 = lim_{x -> 0} [(sin^2 x - x^2) / (x^2 * sin^2 x)].
- Factor the numerator as a difference of squares:
sin^2 x - x^2 = (sin x - x) * (sin x + x).
- Replace sin^2 x in denominator with x^2 using standard limit lim_{x -> 0} (sin x / x) = 1:
x^2 * sin^2 x = x^4 * (sin x / x)^2 ~ x^4.
- Rewrite limit:
L2 = lim_{x -> 0} [(sin x - x) / x^3] * [(sin x + x) / x].
- Evaluate each factor:
Factor 1: lim_{x -> 0} (sin x - x) / x^3:
sin x = x - x^3/6 + x^5/120 - ... => (sin x - x)/x^3 = -1/6 + x^2/120 - ...
lim = -1/6.
Factor 2: lim_{x -> 0} (sin x + x) / x = lim (sin x / x + 1) = 1 + 1 = 2.
- Multiply the two limits:
L2 = (-1/6) * 2 = -2/6 = -1/3.
Conclusion:
(i) Limit = 1/3.
(ii) Limit = -1/3.

Solution to Question 32: Evaluate the exponential indeterminate limits: (i) lim_{x -> 0} (sin x / x)^(1 / x^2) [Form 1^infinity], (ii) lim_{x -> 0+} x^(sin x) [Form 0^0].

ln L = lim_{x -> 0+} [ln x / csc x] (Form -infinity / infinity).
- Apply L'Hopital's Rule:
d/dx [ln x] = 1/x
d/dx [csc x] = -csc x * cot x = -cos x / sin^2 x.
ln L = lim_{x -> 0+} [(1/x) / (-cos x / sin^2 x)]
= lim_{x -> 0+} [- sin^2 x / (x * cos x)]
= - [lim_{x -> 0+} (sin x / x)] * [lim_{x -> 0+} (sin x / cos x)]
= - (1) * (0 / 1) = 0.
- Exponentiate:
L = e^(ln L) = e^0 = 1.
Conclusion:
(i) Limit = e^(-1/6).
(ii) Limit = 1.

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1. Derivation of Parametric Formula:
- Let x = x(t), y = y(t). Denote derivatives with respect to t by dots or primes:
y1 = dy / dx = (dy/dt) / (dx/dt) = y' / x'.
- Compute second derivative y2 = d^2 y / dx^2:
y2 = d/dx [y' / x'] = (d/dt [y' / x']) / (dx / dt)
= [(x' * y'' - y' * x'') / (x')^2] / x'
= (x' * y'' - y' * x'') / (x')^3.
- Substitute y1 and y2 into the Cartesian curvature formula rho = [1 + y1^2]^(3/2) / |y2|:
1 + y1^2 = 1 + (y'/x')^2 = [(x')^2 + (y')^2] / (x')^2.
[1 + y1^2]^(3/2) = [(x')^2 + (y')^2]^(3/2) / (x')^3.
- Divide by |y2|:
rho = [[(x')^2 + (y')^2]^(3/2) / (x')^3] / [|x' * y'' - y' * x''| / (x')^3]
= [(x')^2 + (y')^2]^(3/2) / |x' * y'' - y' * x''|. (Proved).
2. Radius of Curvature of the Cycloid:
x = a*(theta - sin theta), y = a*(1 - cos theta).
- First derivatives with respect to theta:
x' = a*(1 - cos theta) = 2a * sin^2(theta / 2).
y' = a * sin(theta) = 2a * sin(theta / 2) * cos(theta / 2).
- Second derivatives:
x'' = a * sin(theta).
y'' = a * cos(theta).
- Compute (x')^2 + (y')^2:
(x')^2 + (y')^2 = a^2 * [(1 - cos theta)^2 + sin^2(theta)]
= a^2 * [1 - 2*cos theta + cos^2 theta + sin^2 theta]
= a^2 * [2 - 2*cos theta] = 2*a^2 * (1 - cos theta)
= 4*a^2 * sin^2(theta / 2).
Therefore: [(x')^2 + (y')^2]^(3/2) = [4*a^2 * sin^2(theta / 2)]^(3/2) = 8*a^3 * sin^3(theta / 2).
- Compute (x' * y'' - y' * x''):
x' * y'' = a*(1 - cos theta) * a*cos theta = a^2 * (cos theta - cos^2 theta).
y' * x'' = a*sin theta * a*sin theta = a^2 * sin^2 theta.
x' * y'' - y' * x'' = a^2 * [cos theta - (cos^2 theta + sin^2 theta)] = a^2 * (cos theta - 1)
= - a^2 * (1 - cos theta) = - 2*a^2 * sin^2(theta / 2).
Magnitude = 2*a^2 * sin^2(theta / 2).
- Calculate rho:
rho = [8*a^3 * sin^3(theta / 2)] / [2*a^2 * sin^2(theta / 2)]
= 4a * sin(theta / 2).
Conclusion:
For the cycloid, rho = 4a * sin(theta / 2) = 2 * sqrt(2 * a * y).

Solution to Question 15: >> Recall the polar coordinate formula for radius of curvature: rho = (r^2 + r'^2)^(3/2) / |r^2 + 2*r'^2 - r * r''|. Find rho for the cardioid r = a*(1

= r^2 + (r')^2.
Numerator = [(x')^2 + (y')^2]^(3/2) = [r^2 + (r')^2]^(3/2).
- Second derivatives:
x'' = r'' cos theta - 2 r' sin theta - r cos theta
y'' = r'' sin theta + 2 r' cos theta - r sin theta.
- Denominator term x' * y'' - y' * x'':
Multiplying and using trigonometric identities simplifies to:
x' * y'' - y' * x'' = r^2 + 2*(r')^2 - r * r''.
- Hence: rho = [r^2 + (r')^2]^(3/2) / |r^2 + 2*(r')^2 - r * r''|. (Proved).
2. Radius of Curvature for Cardioid r = a*(1 + cos theta):
- r = a*(1 + cos theta).
- r' = -a * sin theta.
- r'' = -a * cos theta.
- Compute r^2 + (r')^2:
r^2 + (r')^2 = a^2 * (1 + cos theta)^2 + a^2 * sin^2 theta
= a^2 * [1 + 2*cos theta + cos^2 theta + sin^2 theta]
= a^2 * [2 + 2*cos theta] = 2*a^2 * (1 + cos theta)
= 4*a^2 * cos^2(theta / 2).
Numerator = [4*a^2 * cos^2(theta / 2)]^(3/2) = 8*a^3 * cos^3(theta / 2).
- Compute Denominator: r^2 + 2*(r')^2 - r * r'':
r^2 = a^2 * (1 + 2*cos theta + cos^2 theta)
2*(r')^2 = 2*a^2 * sin^2 theta
-r * r'' = - [a*(1 + cos theta)] * [-a * cos theta] = a^2 * (cos theta + cos^2 theta)
Sum = a^2 * [1 + 2*cos theta + cos^2 theta + 2*sin^2 theta + cos theta + cos^2 theta]
= a^2 * [1 + 3*cos theta + 2*(cos^2 theta + sin^2 theta)]
= a^2 * [1 + 3*cos theta + 2] = 3*a^2 * (1 + cos theta)
= 6*a^2 * cos^2(theta / 2).
- Evaluate rho:
rho = [8*a^3 * cos^3(theta / 2)] / [6*a^2 * cos^2(theta / 2)]
= (4/3) * a * cos(theta / 2).
- Express in terms of r:
Since r = 2a * cos^2(theta / 2) => cos(theta / 2) = sqrt(r / (2a)):
rho = (4/3) * a * sqrt(r / (2a)) = (2/3) * sqrt(2 * a * r).

>>Solution to Question 15: Use the pedal formula for the radius of curvature: rho = r * (dr / dp). Apply this formula to find the radius of curvature of the equiangul

= r * (dr / ds) * [dtheta / ds + dphi / ds]
= r * (dr / ds) * (dpsi / ds)
= (r / rho) * (dr / ds).
Divide both sides by dr / ds:
dp / dr = r / rho => rho = r * (dr / dp). (Proved).
2. Application to Equiangular Spiral:
r = a * e^(theta * cot alpha).
- Take natural logarithm: ln r = ln a + theta * cot alpha.
- Differentiate with respect to theta:
(1 / r) * (dr / dtheta) = cot alpha => dr / dtheta = r * cot alpha.
- Relate to angle phi:
tan(phi) = r * (dtheta / dr) = r / (r * cot alpha) = tan alpha => phi = alpha (constant).
- Pedal equation:
p = r * sin(phi) = r * sin(alpha).
- Differentiate p with respect to r:
dp / dr = sin(alpha) => dr / dp = 1 / sin(alpha) = csc(alpha).
- Radius of Curvature:
rho = r * (dr / dp) = r * csc(alpha).
Conclusion:
The radius of curvature of the equiangular spiral is directly proportional to its distance r
from the pole: rho = r * csc(alpha).

>>Solution to Question 17: Derive the coordinates (x_bar, y_bar) of the center of curvature for a curve y = f(x). Hence, find the equation of the circle of curvature at

>>Solution Statement Question Define the evolute of a curve. Derive the Explain the steps involved in the method for finding asymptotes parallel to y^2 = a^2x^2 / (x^2 - a^2)

Arrange f(x, y) in descending powers of y:
y^n * phi_0(x) + y^(n-1) * phi_1(x) + ... = 0.
The asymptotes parallel to the y-axis are obtained by equating to zero the coefficient of the highest power of y:
phi_0(x) = 0 (provided phi_0(x) is not a mere numerical constant).
- Asymptotes parallel to x-axis (Horizontal Asymptotes):
Arrange f(x, y) in descending powers of x:
x^n * psi_0(y) + x^(n-1) * psi_1(y) + ... = 0.
Equate to zero the coefficient of the highest power of x:
psi_0(y) = 0 (provided psi_0(y) is not constant).
3. Oblique Asymptotes (y = mx + c):
- For an algebraic curve of degree n:
f(x, y) = x^n * phi_n(y/x) + x^(n-1) * phi_{n-1}(y/x) + x^(n-2) * phi_{n-2}(y/x) + ... = 0.
- Put x = 1 and y = m in the terms of degree n:
phi_n(m) = 0 gives the slopes m of the asymptotes.
- For each non-repeated root m:
c = - phi_{n-1}(m) / phi'_n(m) (provided phi'_n(m) != 0).
- If m is a double root (phi_n(m) = 0 and phi'_n(m) = 0):
(c^2 / 2!) * phi''_n(m) + c * phi'_{n-1}(m) + phi_{n-2}(m) = 0 gives two parallel asymptotes.

Solution to Question 20: Find all the asymptotes (parallel and oblique) of the cubic curve: x^3 + 3x^2*y - 4y^3 - x + y + 3 = 0.

- phi''_3(m) = d/dm [3 - 12m^2] = -24m.
phi''_3(-1/2) = -24*(-1/2) = 12.
- phi_2(m) = 0 => phi'_2(m) = 0.
- phi_1(-1/2) = -1 + (-1/2) = -3/2.
Substitute into equation:
(c^2 / 2) * 12 + c * 0 + (-3/2) = 0
=> 6 * c^2 - 3/2 = 0
=> 6 * c^2 = 3/2 => c^2 = 3 / 12 = 1/4 => c = +- 1/2.
Asymptotes for m = -1/2:
y = (-1/2)*x + 1/2 => 2y = -x + 1 => x + 2y - 1 = 0.
y = (-1/2)*x - 1/2 => 2y = -x - 1 => x + 2y + 1 = 0.
Conclusion:
The three asymptotes are:
1. x - y = 0
2. x + 2y - 1 = 0
3. x + 2y + 1 = 0.

Solution to Question 21: Determine all asymptotes of the Folium of Descartes: x^3 + y^3 = 3axy. Explain how the curve approaches its asymptote.

quadrant and from the RIGHT in the second quadrant.

- y-intercepts: Set x = 0 and solve for y.
4. Asymptotes:
- Vertical asymptotes: Equate coefficient of highest power of y to zero.
- Horizontal asymptotes: Equate coefficient of highest power of x to zero.
- Oblique asymptotes: Use y = mx + c with phi_n(m) = 0.
5. Region of Existence:
- Solve explicitly for y in terms of x: y = +- g(x).
- Determine intervals of x where g(x) is real (i.e., radicands >= 0).
- If g(x) becomes imaginary, no part of the curve exists in that region.
6. Special Points and Sign of Derivatives:
- Compute dy/dx:
* dy/dx = 0 gives points where tangents are parallel to x-axis (extrema).
* dy/dx -> infinity gives points where tangents are vertical.
- Compute d^2y/dx^2 for concavity and points of inflection.

>>Solution to Question 24: Apply the curve tracing algorithm to thoroughly analyze and sketch the Cissoid of Diocles: y^2 * (2a - x) = x^3 (where a > 0).

>>Solution to Question 26: the curve Cadioid = a^2 * (a - x) = x^2 * (a + x) (where a > 0).

tan(phi) = r / (dr / dtheta).
dr / dtheta = -a * sin theta.
tan(phi) = a * (1 + cos theta) / (-a * sin theta) = [2*cos^2(theta/2)] / [
-2*sin(theta/2)*cos(theta/2)]
= -cot(theta / 2) = tan(pi/2 + theta/2).
Therefore: phi = pi/2 + theta/2.
- At theta = 0: phi = pi/2 (tangent is perpendicular to polar axis).
- At theta = pi: phi = pi (tangent is tangent to initial line).
Conclusion:
The curve is a smooth heart-shaped closed oval with a sharp inward cusp at the pole (r = 0,
theta = pi).

>>Solution Statement Question 28: The First and Second Derivative Tests for Local Extrema of Functions of Several Variables Solution Statement Question 29: Differentiable Functions

E + S <=k1 / k-1=> ES --k2--> E + P,
where E = Free Enzyme, S = Substrate, ES = Enzyme-Substrate Complex, P = Product.
2. Differential Rate Equations:
- d[S] / dt = -k1 * [E] * [S] + k-1 * [ES].
- d[ES] / dt = k1 * [E] * [S] - (k-1 + k2) * [ES].
- d[P] / dt = k2 * [ES] = v (reaction velocity).
3. Conservation of Total Enzyme:
[E]t = [E] + [ES] => [E] = [E]t - [ES].
4. Briggs-Haldane Quasi-Steady-State Assumption (QSSA):
Because enzyme concentration is typically very small compared to substrate ([E]t << [S]), the transient intermediate complex [ES] reaches a rapid balance where d[ES]/dt ~ 0:
k1 * [E] * [S] - (k-1 + k2) * [ES] = 0.
5. Solve for [ES]:
Substitute [E] = [E]t - [ES]:
k1 * ([E]t - [ES]) * [S] = (k-1 + k2) * [ES]
k1 * [E]t * [S] - k1 * [ES] * [S] = (k-1 + k2) * [ES]
k1 * [E]t * [S] = [k-1 + k2 + k1 * [S]] * [ES].
Divide numerator and denominator by k1:
[ES] = [[E]t * [S]] / [(k-1 + k2) / k1 + [S]].
6. Define Constants:
- Michaelis constant: Km = (k-1 + k2) / k1.
- Maximum velocity: Vmax = k2 * [E]t.
7. Final Rate Equation:
v = d[P] / dt = k2 * [ES] = [k2 * [E]t * [S]] / [Km + [S]]
=> v = (Vmax * [S]) / (Km + [S]). (Proved).

>>**Solution to Question 29:** ...el microbial cell growth using the logistic differential equation dX/dt = r * X * (1 - X / K). Solve this non-linear ODE by separation of v

- Second derivative: d^2v / dX^2 = -2r / K < 0 (confirms global maximum).
- Thus, cell biomass growth rate is maximized precisely when biomass reaches HALF the carrying capacity: X = K / 2.
- Maximum growth rate value:

(dX/dt)_max = r * (K / 2) * (1 - 1/2) = r * K / 4.

4. Time to Reach Inflection Point:

Set X(t_inf) = K / 2:

K / [1 + ((K - X0) / X0) * e^(-r * t_inf)] = K / 2

=> 1 + ((K - X0) / X0) * e^(-r * t_inf) = 2

=> e^(-r * t_inf) = X0 / (K - X0) => t_inf = (1 / r) * ln((K - X0) / X0).

>>Solution to Question 30: Analyze substrate inhibition enzyme kinetics governed by the Haldane model: v = Vmax * S / (Km + S + S^2 / Ki). Determine the optima